

# Program 5 Requirements

## Topic: json and the web

Hopefully you already have your Github web site. If not, find the instructions in the *Your Web Site* notes (wk 14). Your program will need to read and display the json data file given on the course website: [cit5students.zip](#) Unzip (right-click and then *extract all*) before using. Once decompressed, the file will be a JSON file named *cit5students.json*.

In the file is an array of roughly 20 JavaScript objects in JSON format. Here's an example of the structure. Each object will have a name, major, midterm score and final exam score.

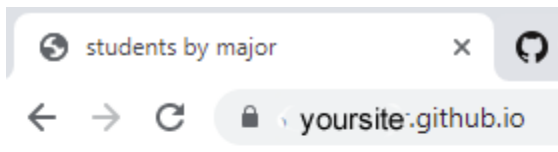
```
[  
  {"name":"Rosie","major":"CIT","midterm":88,"final":88},  
  {"name":"Bruce","major":"BUS","midterm":78,"final":91},  
  ... and so on  
]
```

When first run, your program should initially display a page the looks more or less like this, which offers a display of two buttons. The initial page should be named *index.html*.



Clicking either button will sort and display the json file by major. Clicking the CIT Majors button will display only the CIT majors, as shown on the left, and clicking the Business Majors button should display only the BUS majors, as shown on the right. The names shown below are old data from a previous class, and may not match, necessarily, the names in the data file above, but the result of a button click should look like one these, more or less. Your program should display the data from the *cit5students.json* file given above.

The left image below is the result of clicking the CIT Majors button. The right image is the result of clicking the Business Major button.



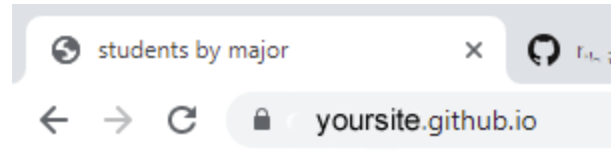
## Select a Major

CIT Majors

Business Majors

### We filter our dataset by major

| Name  | Major | Midterm | Final |
|-------|-------|---------|-------|
| Alice | CIT   | 78      | 87    |
| Aaron | CIT   | 68      | 81    |
| Abner | CIT   | 85      | 77    |
| Agnes | CIT   | 98      | 77    |
| Angie | CIT   | 79      | 64    |
| Artie | CIT   | 89      | 80    |



## Select a Major

CIT Majors

Business Majors

### We filter our dataset by major

| Name  | Major | Midterm | Final |
|-------|-------|---------|-------|
| Abbie | BUS   | 75      | 87    |
| Alfie | BUS   | 53      | 61    |
| Alisa | BUS   | 93      | 92    |
| Alvin | BUS   | 88      | 56    |
| Armin | BUS   | 73      | 77    |
| Chuck | BUS   | 80      | 54    |
| Darla | BUS   | 88      | 61    |

You will need to use a Handlebars template that includes an `#each` section to format the displayed data into an HTML table.

You will need to upload the `index.html` file along with the `cit5students.json` (unmodified) file to your github website so that the program can be run from there.

Your `index.html` page should make an Ajax request that requests the JSON data and then converts the JSON data to a regular JavaScript array of objects. This array of objects should then populate the Handlebars template.

### Grading will follow this approach.

- 100% program runs perfectly with all required items on Github
- 90% program meets all requirement but doesn't always produce the correct output

- 80% program runs without errors, but is missing some requirement.
- 70% program had a good start, most but not all of the code was correct, had one or more fatal errors.
- 40-60% or below. Not very close, many parts missing, would not run, probably a late start.
- 0% no submission or code was not the student's work.

**NOTE:** unless you can get this working on your github website, your maximim score is 70%

When you submit the program for grading you need to do two things:

1. Attach your index.html file to the email just like we have done for earlier programs
2. also your email needs to contain a link to your github website. I should be able to click that link and run your program. The point of this exercise is to get a functioning web site

There are code examples in the notes for "Your Web Page". That is where you should start if you have not already studied those notes.

**Please, please start early on this.**

If you have trouble, attach a copy of the index.html page only to an email, then send it to me.